WHEN THE SIMULATOR FAILS, NOT THE POLICY: AUDITING CONTACT ARTIFACTS IN SIMULATED ROBOT BENCHMARKS

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ABSTRACT

Simulation benchmarks score robot policies by success rate, and every failed episode counts against the policy. Many of these failures are the simulator's fault, not the policy's; we find the cause and fix it. Testing a vision language action model and a world action model in simulation, we find that the policy often places the object and the simulator then ejects it, and that the benchmark's own scripted demonstrations fail the same way. We present SIMUGUARD, an auditing plugin for LIBERO, RoboTwin, RoboCasa and RoboDojo. It leaves the evaluation unchanged. At every physics step, it records the state and flags an object that starts moving at a contact faster than the robot's contact or gravity account for. It replays a recorded episode bit for bit, and it reruns the episode with one simulation parameter changed. With it we trace these episodes on RoboTwin step by step to the collision detection of the physics engine: for two pieces of collision geometry that are one to two centimeters apart, the engine reports a penetration of about two centimeters, and the contact solver removes that depth within the same physics step and throws the object out at several meters per second. The cause is one constant. The engine's distance algorithm discards a triangle whose area falls below a fixed threshold, valid triangles fall below it once the collision geometry is at the centimeter scale, and a later version of the engine lowered the threshold by seven orders of magnitude. With that constant changed and nothing else, the false contacts disappear from the recorded pose sequences, from the benchmark's scripted demonstrations, and from closed loop evaluation, where episodes with an artifact fall from 12 in 100 to none and the success rate on five container tasks rises from 89% to 98%. Success rates in simulation thus count failures that the simulator causes. Benchmarks on PhysX 5.3.1 or earlier should rerun in a repaired engine. Where the engine cannot change, they should report the success rate together with the range it spans when every flagged episode fails or passes: 75.8% to 98.5% around a reported 83.1% for LingBot-VA.

1 INTRODUCTION

Simulation benchmarks give a repeatable way to compare robot policies, from vision language action (VLA) models to world action models (Brohan et al., 2023a;b; Octo Model Team et al., 2024; Kim et al., 2024; Black et al., 2024; Björck et al., 2025; Gemini Robotics Team, 2025; Li et al., 2026). LIBERO (Liu et al., 2023), RoboCasa (Nasiriany et al., 2024) and RoboTwin (Mu et al., 2025; Chen et al., 2025) fix the tasks and the protocol and treat the simulator as faithful physics. Their success rates record which episodes failed, not why. A failure caused by the simulator's treatment of contact enters the same count as a failure caused by the policy. Testing a world action model and a VLA in simulation, we find that such failures are frequent: the policy places the object, and the simulator throws it out.

In an episode of RoboTwin 2.0's Place Can Basket, LingBot-VA (Li et al., 2026) places a can inside the basket, resting against its wall. The gripper releases it. After 0.28 s, within a single 4 ms physics

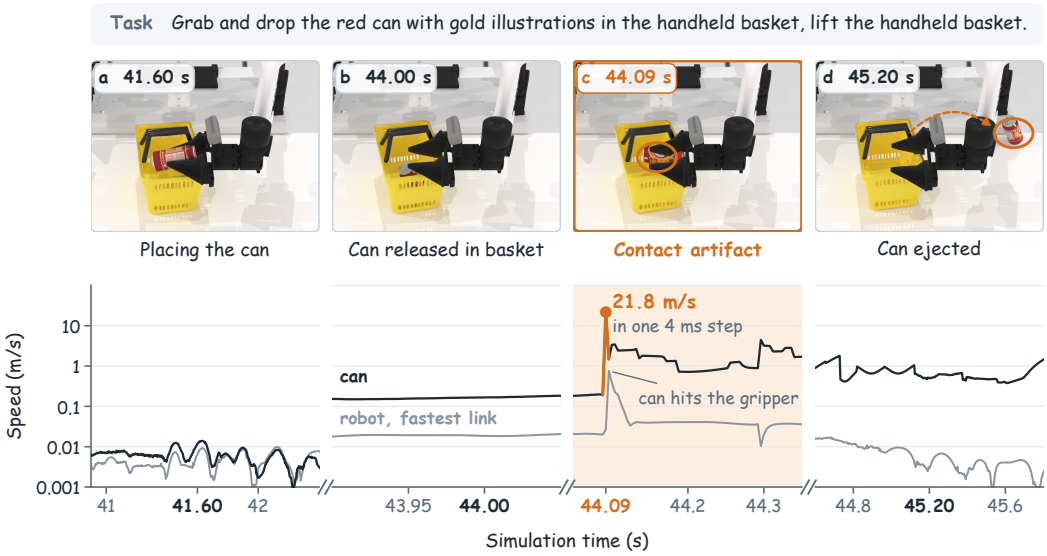


Figure 1: **A contact artifact turns a correct placement into a failure.** After LingBot-VA places the 10 g can in the basket on Place Can Basket (a, b), the can's speed rises from 0.2 to 22 m/s within one 4 ms physics step, a gain of 2.4 J of kinetic energy, while no robot link touches it and the fastest link moves at under 0.1 m/s (c); the can is ejected from the basket and the benchmark records a failure (d). The time axis has four linear segments, one under each frame.

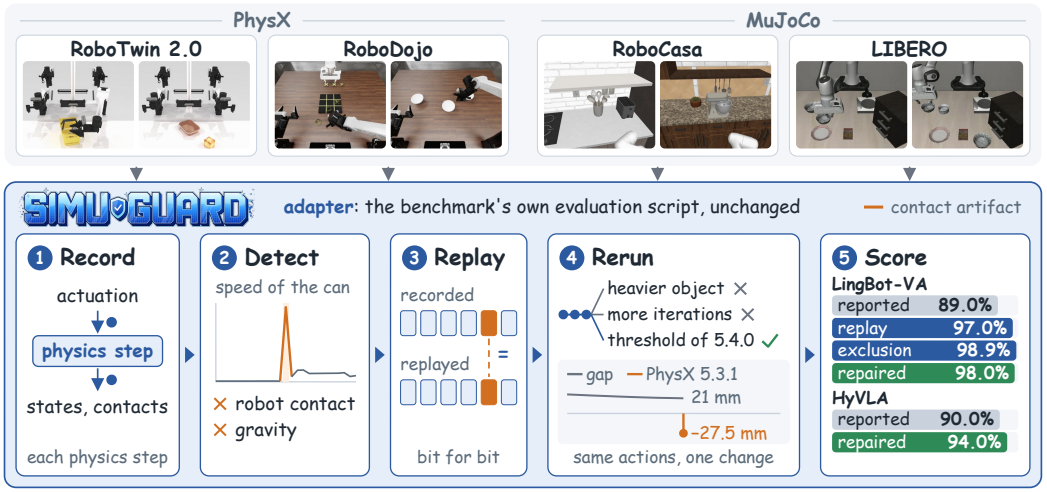


Figure 2: **SIMUGUARD audits a simulated evaluation without changing it.** Attached by an adapter to the benchmark's own evaluation script, SIMUGUARD records every physics step, detects contact artifacts, replays the episode exactly, reruns it with the same actions and one change, and scores the rerun with the benchmark's own success check. Rerun: a heavier object or more solver iterations leave the artifact; the triangle threshold of PhysX 5.4.0 removes it. Inset: the two pieces that start the ejection of Figure 1 are 21 mm apart, and PhysX 5.3.1 reports them 27.5 mm deep; with the threshold of 5.4.0 it reports no contact. Score, five container tasks in closed loop (Table 3): as run (gray), estimated from LingBot-VA's recordings (blue, Section 3.6), and measured in the repaired engine (green).

step, PhysX reports a piece of the can 2.75 cm inside a piece of the basket wall, although the two pieces are 2.1 cm apart, and the can's speed rises from 0.2 to 22 m/s while no robot link touches it and the robot is nearly still; the basket ejects it, the can lands 1.65 m away, and the benchmark records a failure (Figure 1). Nothing in the recording supplies the 2.4 J the can gains: no robot

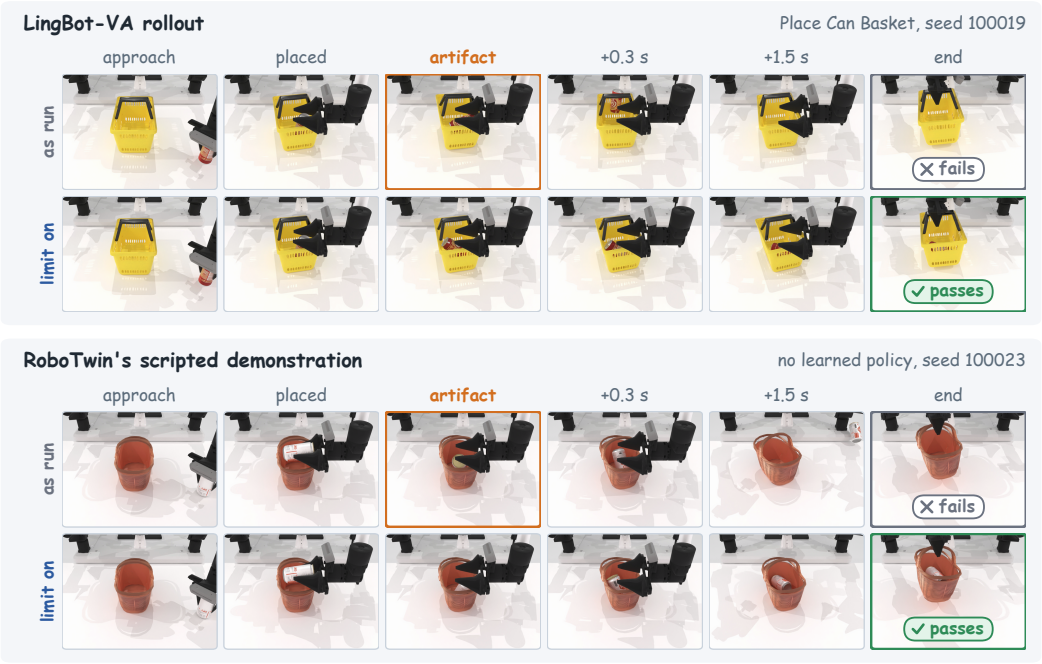


Figure 3: **Two episodes whose outcome the artifact window intervention flips.** Each is shown as run and replayed with the depenetration limit switched on 0.1 s before the first artifact; seed, actions and physics up to the switch are identical. *Top:* LingBot-VA on Place Can Basket. *Bottom:* one of RoboTwin's own scripted demonstrations, which the benchmark discards when it fails.

contact in that step and no drop. Replayed with the same recorded actions and the reported depth removed over several physics steps instead of one, the episode passes the benchmark's own success check (Figure 3); in the engine with one constant of its distance algorithm corrected, the contact is never reported (Section 4.2). We call a contact response that sets an object moving faster than the robot's contact or gravity account for a contact artifact. The impulse that starts one lasts a single physics step and falls between two camera frames; a video shows the can flying, not what launched it, and the success check cannot tell whether the policy or the simulator decided the episode.

SIMUGUARD is an auditing plugin that leaves the evaluation unchanged (Figure 2). An adapter attaches it to the benchmark's own evaluation script; we provide adapters for LIBERO, RoboTwin, RoboCasa and RoboDojo. At every physics step it records the actuation, the object states and the contacts, and flags an object that starts moving at a contact faster than the robot's contact or gravity account for. It replays a recorded episode bit for bit by rebuilding the seeded scene and applying the recorded actuation from the first step. A rerun is a replay with one simulation parameter changed, with the recorded actuation and the success check fixed. Prior work adds metrics to the success rate (Wang et al., 2025; Snyder et al., 2026; Jiang et al., 2026) or measures how solvers depart from physics (Erez et al., 2015; Le Lidec et al., 2023); neither connects an anomalous contact in a rollout to the outcome of that rollout.

In short, many failures in simulated robot benchmarks are the simulator's fault, not the policy's; we find the cause and fix it. Our contributions are:

- **A finding and its cause.** Simulated benchmarks contain frequent episodes in which the policy has placed the object and the simulator then ejects it, so the benchmark scores a failure of the policy. On five RoboTwin tasks that place objects into containers with thin walls, 40 of LingBot-VA's 44 failures and 6 of HyVLA's 10 contain such an ejection, and the benchmark's own scripted demonstrations, which act on ground truth object poses, fail the same way. We trace it to a fixed threshold in the distance algorithm of PhysX 5.3.1, which makes the engine report a penetration between pieces that do not touch; with that constant set to its value in PhysX 5.4.0, the ejections are gone.

- **A tool.** SIMUGUARD monitors every physics step, replays a recorded episode bit for bit, and reruns it with one simulation parameter changed, so that a single contact report can be traced to the outcome of its episode and the trace verified by anyone. It attaches to the benchmark's own evaluation script without changing it, on LIBERO, RoboTwin, RoboCasa and RoboDojo.
- **An evaluation practice.** Where the cause of the artifacts can be removed, we measure the policy in the repaired engine: LingBot-VA succeeds in 98 of 100 closed loop episodes on the five tasks against 89 in the benchmark's engine, with no artifact against 12 episodes. Where a benchmark has to keep its engine, we give the estimates it can compute from its recordings, by ignoring, excluding or replaying the flagged episodes, each under one stated assumption, and compare them with the measurement on the same runs.

2 RELATED WORK

Simulated evaluation of robot policies. VLA policies adapt large pretrained backbones to produce robot actions (Brohan et al., 2023a;b; Open X-Embodiment Collaboration et al., 2024; Octo Model Team et al., 2024; Kim et al., 2024; Black et al., 2024; Björck et al., 2025), and world action models predict future video together with the actions (Du et al., 2023; Wu et al., 2024; Cheang et al., 2024; Hu et al., 2025; Li et al., 2025; 2026). Both are compared on simulated suites built on two families of physics engines: LIBERO (Liu et al., 2023) and RoboCasa (Nasiriany et al., 2024) run on MuJoCo (Todorov et al., 2012), while ManiSkill (Mu et al., 2021; Gu et al., 2023; Tao et al., 2024) and RoboTwin (Mu et al., 2025; Chen et al., 2025) run on SAPIEN (Xiang et al., 2020), which uses PhysX, as does Isaac Lab (Mittal et al., 2023). SimplerEnv (Li et al., 2024b) tests whether their rankings carry over to real robots, and Jiang et al. (2026) audit five of these suites and find shortcut solutions, statistically insignificant gains and overfitting to the test distribution; they ask what a score says about the policy and take the simulator as given. SafeVLA-Bench (Fan et al., 2026) instruments existing benchmarks after the fact to report successes that violate a safety specification, and also takes the simulated physics as correct. All of these suites score each episode with a scripted check on the final state and do not examine the physics that produced it.

Contact simulation. Rigid body engines approximate contact differently. MuJoCo treats it as a soft constraint that removes an overlap over a time constant of several physics steps (Todorov et al., 2012); PhysX treats it as a hard constraint and pushes overlapping bodies apart within the step, up to a configurable maximum speed, and its documentation records that bodies created in overlap may be separated violently and names the depenetration limit as the remedy (NVIDIA Corporation, 2024). Compliant and pressure field models (Castro et al., 2023; Elandt et al., 2019) make other choices. Engines disagree on the same scenes (Erez et al., 2015) and on recorded real impacts (Acosta et al., 2022), contact models trade physical consistency for tractability (Le Lidec et al., 2023), and convex decomposition of assets adds its own approximation (Wei et al., 2022). Collision detection rests on GJK (Gilbert et al., 1988), whose implementations must choose tolerances for termination and degenerate simplices in floating point (van den Bergen, 1999; Ericson, 2004) and whose convergence recent work accelerates (Montaut et al., 2022; 2024); the defect we trace is such a tolerance. These studies compare engines in isolated scenes, document a known behavior or improve the algorithm; none follows one contact report inside a fixed benchmark to the outcome of the episode, which is what SIMUGUARD does (Section A).

Reliability of benchmark numbers. Several studies ask what a success rate measures and how many trials it needs (Kress-Gazit et al., 2024; Vincent et al., 2024). RoboEval (Wang et al., 2025) adds behavioral metrics because equal success rates can hide different executions; Snyder et al. (2026) replace binary outcomes with graded progress; LIBERO-PRO (Zhou et al., 2025) and LIBERO-Plus (Fei et al., 2025) show that high scores often do not survive changes to objects, layouts and observations; and Henderson et al. (2018), Agarwal et al. (2021) and Bouthillier et al. (2021) show how variance and point estimates distort comparisons. These works improve how outcomes are aggregated and take the outcome of each episode as given. Developers who test robots in simulation name unrealistic physics among their main obstacles (Afzal et al., 2021). We test the outcome itself: SIMUGUARD traces one contact report to the outcome of the episode that contains it, with a replay anyone can verify, and measures the score once the cause is removed.

3 SIMUGUARD

SIMUGUARD has three requirements: it must not change what it measures, it must reproduce an episode exactly so that a difference after an intervention belongs to the intervention, and it must run with the benchmark's own evaluator so that its findings apply to the reported numbers.

3.1 SETTING

A benchmark evaluates a policy on a sequence of seeds. Each seed fixes the scene, and the evaluator runs the policy's actions through the simulator until a scripted success check passes or the action budget is spent. The simulator advances in physics steps of a few milliseconds (4 ms in RoboTwin, 2 ms in LIBERO and RoboCasa); one policy action spans tens of them. Of N scored episodes, S succeed, and the benchmark reports S/N.

A contact artifact is motion that starts at a contact and that neither the robot's contact nor gravity accounts for; in this paper the term names what the detector of Section 3.3 flags, and Section 4.5 measures its precision against human judgment. Let A be the set of scored episodes that contain one. Figure 2 shows the five steps that follow.

3.2 RECORDING WITHOUT INTRUSION

SIMUGUARD wraps the physics step of the running scene, so the evaluator, the benchmark and the policy run unmodified in their own process. A thin adapter per engine provides the wrapper, around the step function of SAPIEN (RoboTwin), Isaac Sim (RoboDojo) or robosuite, which runs LIBERO and RoboCasa on MuJoCo. Before each step, SIMUGUARD records the actuation: joint drive targets and forces in PhysX; control inputs, applied forces and mocap poses in MuJoCo. It also records any state that benchmark code writes between two steps, which it must replay (Section 3.4). After each step, it records the pose and velocity of every body and the contacts, merged per pair of bodies, with impulse and penetration depth. A task specification names the objects a task manipulates and the containers they go into. The evaluator, the benchmark and the policy code are untouched, and with the monitor on or off, all 59 tracked object trajectories in our check are identical; the monitor adds 7.5 to 11.6 ms to each 4 ms step in SAPIEN (Section C.1).

3.3 DETECTING CONTACT ARTIFACTS

Detection has three stages, with the same thresholds for every task (Section B.5). The first finds motion that starts at a contact. An object becomes a candidate when it touched another body within the last 40 ms, moves faster than 0.5 m/s, and changes velocity by at least 0.35 m/s within one 4 ms step. The candidate is confirmed if within 240 ms it travels at least 4 cm and either flies free for 16 ms or stays in violent contact with its container.

The second stage removes motion that gravity explains. An object dropped into a bin also moves fast right after a contact. For each confirmed event, SIMUGUARD compares the object's peak speed in the 0.1 s after onset with the free fall speed $\sqrt{2g\,\Delta z}$ for the height it dropped before that peak, and keeps the event only if the ratio exceeds 1.5.

The third stage removes motion that the robot transmits through another body. A basket that the robot's base shoves across the floor carries the bread inside it at the basket's speed. SIMUGUARD finds every body that touched both the object and a robot link in the 40 ms before onset, and discards the event if the object's peak speed is less than 1.5 times the fastest such carrier's speed before onset. An event that passes all three stages is a candidate contact artifact. The stages compare speeds, not the work done on the object, so human review is the final standard: raters judged the flagged clips frame by frame (Section 4.5).

3.4 EXACT REPLAY

SIMUGUARD replays an episode by rebuilding the scene from its seed and applying the recorded actuation from the first physics step. The replay matches the recording bit for bit (Section 4.5); three details make this hold. PhysX cannot restore a saved state exactly once bodies touch, since trajectories restored through its state interface diverge by millimeters to centimeters, so

SIMUGUARD always replays from the start. Benchmark code changes the state between physics steps: a RoboCasa fixture sets its door joint, and robosuite 1.4, which LIBERO uses, recomputes the dynamics before each step and so rewrites the solver's warm start; SIMUGUARD records both and writes them back. Finally, outcomes depend on values outside the physics: RoboTwin's success checks read the gripper opening the robot was commanded to, which SIMUGUARD restores at each step from the recorded drive targets.

3.5 CONTROLLED RERUNS

A rerun is a replay with one simulation parameter changed: the mass of the manipulated object, the number of solver iterations, the maximum speed at which PhysX separates overlapping bodies (its maximum depenetration velocity), or the time constant of MuJoCo's contacts. A rerun either applies the change from the first physics step, or switches it on 0.1 s before the episode's first contact artifact. In the second form, everything before the switch is identical to the recording, which SIMUGUARD verifies bit for bit; from the switch on, the limit acts on every body until the episode ends. A different outcome therefore comes from the physics after the switch, of which the artifact is the first event the recording shows: an artifact window intervention, not an intervention on one contact. The benchmark's own success check scores every rerun, and the evaluator stops at the first success, as in the original evaluation. A third form keeps the change on for a new closed loop rollout on the same seed; the policy samples new actions, so the episode is no longer comparable one by one with the recording. The change can also be in the engine's code (Section 4.2).

3.6 SCORING

Once the flagged episodes A of an evaluation are known, and if only they can change, the success rate under a physics that never ejects lies between $(S - S_A)/N$, every flagged episode failing, and $(S - S_A + |A|)/N$, every flagged episode succeeding, where S_A counts the successes in A; the one assumption, that no unflagged episode changes, fails where the detector misses an artifact. On the five RoboTwin tasks of Section 4.1 it is $[75.8\%, 98.5\%]$ for LingBot-VA and $[86.0\%, 96.0\%]$ for HyVLA. Each of the four rates we report is a point inside it under one further assumption (Figure 2, Score). The *reported* rate S/N assumes that every ejection is the policy's fault. The *replay* rate replaces the recorded outcome $y_e \in \{0, 1\}$ of each flagged episode by the outcome r_e of its recorded actions under the limit switched on before the artifact,

$$\hat{s} \;=\; \frac{1}{N}\Big(S \;+\; \sum_{e \in A}(r_e - y_e)\Big), \tag{1}$$

and assumes that those actions remain the policy's choice once the ejection is removed. The *exclusion* rate $(S - S_A)/(N - |A|)$ assumes that the flagged episodes are as hard as the others. The *closed loop* rate is the success rate of new rollouts with the limit kept on, and assumes that the limit changes nothing the task needs. All four keep the benchmark's engine. Where the cause of the artifacts is removed from the engine, the success rate of new rollouts in the repaired engine measures the rate without the artifacts; Section 4.3 reports it next to the estimates.

4 EXPERIMENTS

Setup. We run RoboTwin 2.0 (Chen et al., 2025) through its official evaluation script in its clean configuration (no background, table or lighting randomization, seen instructions), on SAPIEN (Xiang et al., 2020) with PhysX and the benchmark's 4 ms physics step. We audit two kinds of policy, each with its RoboTwin checkpoint: LingBot-VA (Li et al., 2026), a world action model that predicts future video together with its actions, on all 50 tasks, and HyVLA (Zhang et al., 2026), a vision language action model, on the five container tasks of Section 4.1 and six other tasks. Before the policy runs on a seed, RoboTwin runs a scripted demonstration and scores the policy only on seeds where the demonstration passes; every rate in this paper is over scored episodes, and skipped seeds do not enter N. On MuJoCo, with a 2 ms physics step, we run $\pi_{0.5}$ (Physical Intelligence et al., 2025) on the 40 tasks of LIBERO and XR-1 (Fan et al., 2025) on all 50 tasks of its RoboCasa evaluation set. RoboDojo (Chen et al., 2026) runs on Isaac Sim with PhysX and a 4 ms physics step; we run XR-1 on four of its tasks. SIMUGUARD monitors every episode, and all analysis uses the stored recordings. Episode counts, versions and checkpoints are listed in Section B.

Table 1: **Success rates on RoboTwin 2.0** under the four scores of Section 3.6, in percent. *Rep.*: as run. *Replay*: Equation (1), every episode with an artifact replayed with the limit switched on before it. *Excl.*: rate over the episodes without an artifact. *CL*: new closed loop rollouts with the limit on; as run rate on the shared seeds 81.8% (LingBot-VA) and 90.7% (HyVLA). Counts and the tasks outside the five: Table 22.

	LingBot-VA (world action model)				HyVLA (vision language action)			
Task	Rep.	Replay	Excl.	CL	Rep.	Replay	Excl.	CL
Place Can Basket	74.4	86.7	96.2	90.0	80.0	75.0	81.3	80.0
Stack Bowls Three	76.0	90.0	100.0	100.0	90.0	95.0	100.0	100.0
Place Object Basket	87.5	90.0	94.1	50.0	95.0	95.0	95.0	45.5
Place Bread Basket	97.5	100.0	100.0	85.0	100.0	100.0	100.0	70.0
Stack Bowls Two	92.5	100.0	100.0	100.0	85.0	100.0	100.0	100.0
Five tasks	83.1	91.9	98.0	87.1	90.0	93.0	95.6	78.4

Table 2: **One constant.** *False*: the 20 onset contacts we traced (4 on Place Can Basket, 16 on Stack Bowls Three) whose two pieces do not intersect, recomputed outside the simulator from the recorded poses. *Real*: 4 onset contacts whose pieces do intersect. *Expert*: RoboTwin's scripted demonstration on 60 seeds per task inside the simulator, episodes with an artifact and episodes that fail after a valid plan.

	Contacts		Expert, artifact		Expert, failed	
Engine	False (20)	Real (4)	Can	Bowls	Can	Bowls
PhysX 5.3.1 (RoboTwin)	20	4	18	6	6	5
PhysX 5.3.1, threshold of 5.4.0	0	4	1	0	1	0
PhysX 5.9.0	0	4	–	–	–	–

4.1 Where contact artifacts occur

Failures with contact artifacts concentrate in five tasks that place an object into a container with thin walls (Table 1). Of LingBot-VA's 44 failures on these tasks, 40 contain an artifact; in its other 45 tasks, 12 of 146 failures do, on eight tasks (Table 23). HyVLA shows the same pattern: 6 of its 10 failures on the five tasks contain an artifact, and none of its 38 failures on the six other tasks.

An artifact does not always end in failure: 17 successful LingBot-VA episodes of the five tasks contain one, 16 of them on Place Can Basket, where an ejected can may land back in the basket. Artifacts also act before the policy is scored: on Place Can Basket, 16 of the 22 seeds RoboTwin skipped because its own scripted demonstration failed contain one, and 13 pass with the limit (Figure 3, Section D.3).

4.2 Cause: a fixed threshold in the collision detection

At the onset of an artifact, PhysX reports a contact that the geometry does not have. Over the 99 artifacts on Place Can Basket, the reported overlap between can and basket at the onset step is 21.5 mm in the median. In the 35 episodes whose first artifact starts with such a report, the two convex pieces that report it are a median 15 mm apart when the step begins; the reported normal points from the can into the basket in 34, the opposite of its direction in all 1958 other contacts between the two bodies one step earlier; and the reported depth is the distance from the far side of the can's piece to the contact point, that is, the gap plus the thickness of the piece (median difference below 0.01 mm). The can moves under 1 mm in that step.

The error is in the engine's distance computation. RoboTwin runs on SAPIEN 3.0.0b1, which contains PhysX 5.3.1. Its GJK algorithm (Gilbert et al., 1988) discards a triangle abc of its simplex as degenerate when $|ab \times ac|^2 \leq 2^{-23}$, a fixed number in units of meters that does not scale with the geometry. RoboTwin decomposes its containers into convex pieces one to two centimeters across, and at the four onsets we traced, the value lies between 6.0×10^{-8} and 1.189×10^{-7}, below the threshold of 1.192×10^{-7}. The algorithm then stops before it has found the closest points, the

Table 3: **Closed loop evaluation in the repaired engine.** Successes of 20, in parentheses episodes with an artifact. *Bench.*: RoboTwin as released (LingBot-VA reseeded per episode; HyVLA: the run of Table 1). *Offset*: contact offset 1 mm. *Repaired*: threshold of PhysX 5.4.0.

	LingBot-VA			HyVLA	
Task	Bench.	Offset	Repaired	Bench.	Repaired
Place Can Basket	15 (5)	19 (3)	20 (0)	16 (4)	18 (0)
Stack Bowls Three	15 (6)	17 (3)	19 (0)	18 (3)	20 (0)
Place Object Basket	20 (0)	20 (0)	20 (0)	19 (0)	18 (0)
Place Bread Basket	19 (1)	19 (0)	20 (0)	20 (0)	18 (1)
Stack Bowls Two	20 (0)	20 (0)	19 (0)	17 (3)	20 (0)
Five tasks, of 100	89 (12)	95 (6)	98 (0)	90 (10)	94 (1)

contact generation picks a face that points away from the other piece, and the gap is reported as a depth (Section D.1). PhysX 5.4.0 squares the threshold. A program that feeds the recorded poses of one pair of pieces to PhysX's contact generation outside the simulator returns the contact the simulator reported, digit for digit, for all 20 false contacts we traced (Table 2). With the threshold of 5.4.0 and no other change, none of the 20 is reported, and 4 contacts in which the pieces do intersect keep their depth. The same change inside the simulator removes the artifacts from the benchmark's own scripted demonstrations: over 60 seeds, the episodes with an artifact fall from 18 to 1 on Place Can Basket and from 6 to 0 on Stack Bowls Three, and the demonstrations that fail after a valid plan from 6 to 1 and from 5 to 0.

Three settings turn the false contact into an ejection without causing it (Sections D.1 and D.2). Every shape generates contacts from 10 mm away, so the contact report of can and basket holds 25 to 90 pairs of pieces at each step; with this offset at 1 mm from just before the artifact, 4 of the 37 Place Can Basket episodes keep one, and 6 of 12 on Stack Bowls Three, where the pieces touch. The engine removes a reported depth at any speed; limited to 1 m/s, the object leaves more slowly, but 26 of the 37 episodes keep an artifact. And the objects weigh 10 g.

4.3 What the artifacts do to the score

In the repaired engine, LingBot-VA succeeds in 98 of 100 closed loop episodes on the five tasks, against 89 in the benchmark's engine (Table 3). In the benchmark's engine, 12 episodes contain an artifact and 10 of the 11 failures are among them; in the repaired engine no episode contains one. Of the 83 seeds scored in both arms, 8 succeed only in the repaired engine and 2 only in the benchmark's (sign test $p = 0.11$). The scripted demonstration, which decides the seeds a policy is scored on, fails on 49 seeds in the benchmark's engine and on 36 in the repaired one. With the contact offset at 1 mm and the engine unchanged, 95 episodes succeed and 6 keep an artifact. HyVLA succeeds in 94 episodes in the repaired engine against 90 in its audited run, with 1 episode with an artifact against 10; in that one a bread leaves another bread at 2.4 m/s, a case we have not traced.

A benchmark that keeps its engine can only estimate this rate from its recordings (Section 3.6). On the benchmark arm above, excluding the 12 flagged episodes gives 87 of 88 (98.9%), and replaying them with the depenetration limit gives 97 of 100, since 8 of the 10 failures pass and both successes are kept. On the larger audited runs of Table 1, whose 59 replays all match their recordings bit for bit up to the switch, the replay rate is 91.9% against 83.1% reported and 98.0% under exclusion for LingBot-VA, losing 9 of its 19 successes with an artifact, and 93.0% against 90.0% and 95.6% for HyVLA. A limit kept on for whole rollouts changes the task: Place Object Basket falls from 90.0% to 50.0% under it (*CL*) and stays at 20 of 20 in the repaired engine.

4.4 Other platforms

SimuGuard also attaches to RoboDojo, LIBERO and RoboCasa, audited at a smaller scale (Table 4). RoboDojo runs Isaac Sim with contact generation on the GPU, and 7 of its 40 XR-1 episodes contain artifacts, all failures. In the one we traced, the engine reports a tube 11 mm inside a wall of its rack at a point 7.6 mm outside the tube's convex hull: a false contact again, on a path the threshold above does not enter and whose cause we have not located (Section E). The depenetration

Table 4: **Four benchmarks, two engines.** Episodes with a contact artifact, the fastest object at an artifact, and its speed replayed with the depenetration limit of 1 m/s (RoboTwin: median, Place Can Basket; RoboDojo: Figure 6).

Benchmark	Engine	Policy	Episodes	With artifact	Peak (m/s)	Limited (m/s)
RoboTwin 2.0	PhysX, SAPIEN	LingBot-VA	2060	80	21.8	1.39
		HyVLA	220	10	–	–
RoboDojo	PhysX, Isaac Sim	XR-1	40	7	6.30	1.88
LIBERO	MuJoCo	$\pi_{0.5}$	400	2	1.57	–
RoboCasa	MuJoCo	XR-1	300	5	1.19	–

limit of 1 m/s slows that tube from 4.31 to 1.88 m/s. On MuJoCo, artifacts are few and slow: 2 of 400 LIBERO episodes and 5 of 300 RoboCasa episodes, none above 1.6 m/s.

4.5 Reliability of replay and detection

SimuGuard reproduces every recorded episode exactly, and only with the replay details of Section 3.4 (Table 8). Detection agrees with human judgment: five robotics engineers rated each clip independently, without knowing where it came from, and settled disagreements by discussion (Table 8, Section G). On Place Can Basket, they judged 37 of 40 clips centered on detected events physically implausible (precision 0.925) and none of 20 hard negatives.

5 Discussion and Limitations

Discussion. The ejections on RoboTwin have one cause in the engine, and removing it removes them for a single pair of collision pieces, in the scripted demonstrations and in closed loop evaluation (Tables 2 and 3). The cause needs two conditions that RoboTwin meets together: PhysX 5.3.1 or earlier, which the SAPIEN releases we inspected (3.0.0b1, 3.0.0 and 3.0.3) contain, and collision geometry at the centimeter scale. It does not belong to a learned policy: the scripted expert fails in the same way and stops failing in the repaired engine. Slowing the removal of the reported depth is not a repair, since the limit kept on for whole rollouts halves the rate of a task that the repaired engine leaves unchanged. For a benchmark this suggests an order of work: record every step, flag the artifacts, trace a recurring one to its cause, and measure in the repaired engine; estimates from recordings serve where the engine cannot change, and on our runs they bracket the measurement.

Limitations. The closed loop measurement in the repaired engine covers two policies on five tasks with 20 seeds each, and the difference in success on LingBot-VA's 83 shared seeds is not significant on its own ($p = 0.11$), unlike the difference in artifacts. We changed one constant inside PhysX 5.3.1 and did not run RoboTwin on a complete later version, in which the steps after the faulty distance are unchanged. On Stack Bowls Three, 4 of the 21 deep contacts we traced are real penetrations and one is not reproduced outside the simulator. The cause of the false contact on RoboDojo is not located, the MuJoCo benchmarks were audited at a smaller scale, and the detector's precision was measured on one task.

6 Conclusion

Success rates in simulation count failures that the simulator causes. On five RoboTwin tasks that place objects into containers with thin walls, 40 of LingBot-VA's 44 failures and 6 of HyVLA's 10 contain a contact artifact, and the benchmark's own scripted demonstrations fail the same way. SimuGuard traces these artifacts to one constant in the collision detection of PhysX 5.3.1. With that constant set to its value in PhysX 5.4.0, the artifacts disappear: closed loop episodes with an artifact fall from 12 of 100 to none, and LingBot-VA's success rate on the five tasks rises from 89% to 98%. Benchmarks on PhysX 5.3.1 or earlier should rerun in a repaired engine. Where the engine cannot change, they should report the range that SimuGuard computes from their recordings next to the success rate, not the success rate alone.

AI USE STATEMENT

In this work, we have not used generative AI tools to generate synthetic data, to develop theoretical models, to formulate or prove mathematical claims, or to propose hypotheses, and translation, dataset cleaning and qualitative analysis are not applicable to this work. We used generative AI tools to create and edit software code, to draft and edit parts of the paper, to identify relevant literature and to format references. We have reviewed all AI-assisted work: every number in the paper comes from runs the authors executed and can be regenerated by exact replay. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

ETHICS STATEMENT

This work studies the validity of simulated robot benchmarks. It involves no personal data; the human judgments in Section G are annotations of simulation videos by five robotics engineers.

REPRODUCIBILITY STATEMENT

SIMUGUARD runs the benchmark's own evaluation script, and every episode in this paper is stored with its seed and the actuation of every physics step, so each result can be regenerated by exact replay. Section B lists tasks, episodes, upstream versions and checkpoints; Section B.5 lists every detector parameter; Section C.1 describes the adapters and the replay; Section D.2 specifies the stress test; Section D.1 describes the program that reproduces the false contact outside the simulator, which the repository contains with the poses of the episode of Figure 1. The code, the detector, the replay and rerun tools, the episode manifest of Section F and the scripts that produce each table are at `https://github.com/ALEXander22568/simuguard`; the recordings will be released with the code.

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A Extended Related Work

Policies. Language conditioned manipulation moved from grounding language in affordances (Ahn et al., 2022) and multimodal prompts (Jiang et al., 2023) to transformers trained on robot data at scale (Brohan et al., 2023b; Open X-Embodiment Collaboration et al., 2024; Octo Model Team et al., 2024) and to VLAs that start from a vision language backbone (Driess et al., 2023; Brohan et al., 2023a; Li et al., 2024a; Kim et al., 2024; Black et al., 2024; Physical Intelligence et al., 2025; Björck et al., 2025; Gemini Robotics Team, 2025). Later work changes the action head and its training (Chi et al., 2023; Zhao et al., 2023; Ke et al., 2024; Liu et al., 2025; Pertsch et al., 2025; Kim et al., 2025) or the spatial input (Zheng et al., 2025; Qu et al., 2025), and large real robot datasets support this training (Walke et al., 2023; Khazatsky et al., 2024). World action models add a generative model of future observations: video prediction as a planner (Du et al., 2023; Ko et al., 2024), video generative pretraining (Wu et al., 2024; Cheang et al., 2024; Hu et al., 2025), a single model for video and actions (Li et al., 2025; 2026), and latent actions learned from video (Ye et al., 2025). They draw on world models trained for control (Wu et al., 2022; Hafner et al., 2025) and on interactive video models (Yang et al., 2024; Bruce et al., 2024). Every one of these policies is scored in simulation by the outcome of an episode; SimuGuard audits that outcome.

Simulated benchmarks. Simulated suites for manipulation span RLBench (James et al., 2020), Meta-World (Yu et al., 2019), robosuite (Zhu et al., 2020), CALVIN (Mees et al., 2022), ManiSkill (Mu et al., 2021; Gu et al., 2023; Tao et al., 2024), LIBERO (Liu et al., 2023), RoboCasa (Nasiriany et al., 2024), RoboTwin (Mu et al., 2025; Chen et al., 2025) and RoboDojo (Chen et al., 2026), and household suites such as iGibson (Li et al., 2021), Habitat (Szot et al., 2021) and BEHAVIOR (Li et al., 2022); GPU physics (Makoviychuk et al., 2021; Mittal et al., 2023) makes large evaluations cheap. Generalization suites vary objects, layouts, lighting and instructions (Pumacay et al., 2024; Garcia et al., 2024; Zhou et al., 2025; Fei et al., 2025), SimplerEnv (Li et al., 2024b) and RoboArena (Atreya et al., 2025) ask whether simulated or crowd sourced rankings agree with real robots, and Acosta et al. (2022) validate engines on recorded impacts. All of them treat the physics engine as part of the fixed apparatus.

Evaluation practice. Guidance on evaluating robot policies concerns the number of trials, the choice of conditions and the reporting of uncertainty (Kress-Gazit et al., 2024; Vincent et al., 2024; Snyder et al., 2026); RoboEval (Wang et al., 2025) and Jiang et al. (2026) examine what the scores of manipulation benchmarks measure; and the machine learning literature documents how seeds and variance distort comparisons (Henderson et al., 2018; Agarwal et al., 2021; Bouthillier et al., 2021). These works improve how outcomes are aggregated. SIMUGUARD asks whether an individual outcome is the policy's.

Contact simulation. Rigid body engines differ in how they model contact: soft constraints with a time constant in MuJoCo (Todorov et al., 2012), hard constraints with penetration correction in PhysX, compliant and pressure field models (Castro et al., 2023; Elandt et al., 2019), and differentiable formulations (Howell et al., 2022). Comparisons across engines (Erez et al., 2015; Le Lidec et al., 2023) and against real impacts (Acosta et al., 2022) report that the same scene produces different forces and trajectories, and convex decomposition of assets adds another approximation (Wei et al., 2022). SIMUGUARD measures what one such difference, the speed at which PhysX removes an overlap, does to the score of a benchmark.

B SETUP

B.1 ROBOTWIN

We use RoboTwin commit 6dde571, XPolicyLab commit fa431ec, LingBot-VA code 7c6ffa9, and the RoboTwin checkpoint of LingBot-VA from ModelScope, driven through its end effector action path. Inference runs on RTX 4090 or H800 GPUs; the copies of the weights are byte identical (sha256 over all 16 files), and launch settings are the same everywhere. RoboTwin runs its scripted demonstration on each seed in turn and skips the seed if the demonstration fails its check, until N seeds are scored. Every run uses RoboTwin's *demo_clean* task configuration, seen instructions, seed base 0 and the expert check. The protocol is 40 episodes per task: the 40 tasks outside the ten of the cross task study were evaluated in two passes of 20 seeds from the same seed base, the second on H800 GPUs with RoboTwin commit 2a14726 and XPolicyLab fe71eb5, the same checkpoint and the same evaluator arguments; the two passes score 675 seeds in common and agree on the outcome of 629 of them. Only the first pass was replayed. Of the ten cross task tasks, Place Can Basket has 90 episodes and Stack Bowls Three 50; the other eight have 40 each, in two passes from the same seed base on RTX 4090 GPUs, both passes replayed; Put Object Cabinet's second pass starts at seed 100026 and needs 199 skipped seeds for its 30, and its three replays diverge from the recording before the switch, so they are listed in Table 23 but not counted. Place Can Basket was evaluated three times with 30 scored seeds each, because a single repetition is not stable: 15 of the 37 seeds involved succeed in some repetitions and fail in others (23, 24 and 20 successes, $74.4 \pm 6.9\%$). Table 5 lists every task.

Table 5: **RoboTwin 2.0 runs** of LingBot-VA. *Artifact*: episodes with a contact artifact among failures and successes. *Skipped*: seeds discarded because the scripted demonstration failed. *Succ.*: successes; *Rep.*: reported rate. Bold: the five tasks of Table 1. The 40 tasks outside the ten of the cross task study were run twice with 20 seeds each, and both passes are pooled.

				Artifact in							Artifact in		
Task	N	Succ.	Rep. (%)	fail.	succ.	Skipped	Task	N	Succ.	Rep. (%)	fail.	succ.	Skipped
Adjust Bottle	40	40	100.0	0	0	11	**Place Can Basket**	3×30	67	74.4	21	16	22
Beat Block Hammer	40	39	97.5	0	0	19	Place Cans Plasticbox	40	40	100.0	0	0	0
Blocks Ranking RGB	40	39	97.5	0	0	0	Place Container Plate	40	39	97.5	0	0	9
Blocks Ranking Size	40	36	90.0	0	0	3	Place Dual Shoes	40	37	92.5	3	2	27
Click Alarmclock	40	40	100.0	0	0	8	Place Empty Cup	40	40	100.0	0	0	4
Click Bell	40	40	100.0	0	0	0	Place Fan	40	36	90.0	1	0	8
Dump Bin Bigbin	40	39	97.5	1	0	36	Place Mouse Pad	40	39	97.5	0	0	0
Grab Roller	40	40	100.0	0	0	2	**Place Object Basket**	40	35	87.5	3	3	16
Handover Block	40	39	97.5	0	0	12	Place Object Scale	40	36	90.0	0	0	16
Handover Mic	40	38	95.0	0	0	6	Place Object Stand	40	40	100.0	0	0	1
Hanging Mug	40	9	22.5	1	0	7	Place Phone Stand	40	39	97.5	0	0	20
Lift Pot	40	40	100.0	0	0	56	Place Shoe	40	39	97.5	0	0	3
Move Can Pot	40	37	92.5	0	0	4	Press Stapler	40	33	82.5	0	0	4
Move Pillbottle Pad	40	40	100.0	0	0	21	Put Bottles Dustbin	40	29	72.5	0	0	15
Move Playingcard Away	40	40	100.0	0	0	0	Put Object Cabinet	40	30	75.0	1	2	272
Move Stapler Pad	40	27	67.5	2	0	2	Rotate QRcode	40	37	92.5	0	0	3
Open Laptop	40	40	100.0	0	0	0	Scan Object	40	38	95.0	0	0	80
Open Microwave	40	31	77.5	0	0	32	Shake Bottle	40	40	100.0	0	0	16
Pick Diverse Bottles	40	38	95.0	0	0	46	Shake Bottle Horizontally	40	40	100.0	0	0	16
Pick Dual Bottles	40	40	100.0	0	0	2	Stack Blocks Three	40	35	87.5	2	0	3
Place A2B Left	40	40	100.0	0	0	9	Stack Blocks Two	40	40	100.0	0	0	0
Place A2B Right	40	37	92.5	0	0	9	**Stack Bowls Three**	50	38	76.0	12	0	43
Place Bread Basket	40	39	97.5	1	0	9	**Stack Bowls Two**	40	37	92.5	3	0	15
Place Bread Skillet	40	39	97.5	0	0	56	Stamp Seal	40	38	95.0	1	0	18
Place Burger Fries	40	40	100.0	0	5	0	Turn Switch	40	21	52.5	0	0	7

All 50 tasks: N = 2060, successes 1870 (90.8%), artifacts in 52 failures and 28 successes, 968 seeds skipped.

HyVLA. We use the RoboTwin checkpoint Hy-Embodied-0.5-VLA-RoboTwin through XPolicyLab, with the deployment settings of its own RoboTwin evaluation (end effector actions, seven actions executed per query), and the same evaluator arguments as for LingBot-VA (seed base, seen instructions, scripted demonstration check), so both policies are scored from the same seed sequence. Place Object Basket ran as five parallel shards that start at different seeds. Because RoboTwin's scripted demonstration can pass on a seed in one run and fail on it in another, two seeds were scored by two shards; we keep one run per seed, which leaves 20 distinct seeds (Table 6).

Table 6: **RoboTwin 2.0 runs** of HyVLA. *Artifact*: failures that contain a contact artifact. *Skipped*: seeds discarded because the scripted demonstration failed.

Task	N	Successes	Reported (%)	Artifact	Skipped
Place Can Basket	20	16	80.0	1/4	5
Stack Bowls Three	20	18	90.0	2/2	18
Place Object Basket	20	19	95.0	0/1	8
Place Bread Basket	20	20	100.0	0/0	5
Stack Bowls Two	20	17	85.0	3/3	7
Blocks Ranking Size	20	17	85.0	0/3	2
Open Microwave	20	9	45.0	0/11	2
Put Bottles Dustbin	20	15	75.0	0/5	10
Put Object Cabinet	20	16	80.0	0/4	156
Stamp Seal	20	15	75.0	0/5	9
Turn Switch	20	10	50.0	0/10	3

B.2 LIBERO

We use upstream LIBERO with robosuite 1.4.0 and MuJoCo 3.2.3, a 2 ms physics step and 25 physics steps per action. The policy is the $\pi_{0.5}$ checkpoint RLinf-Pi05-LIBERO-130-fullshot-SFT (Yu et al., 2025), evaluated with the protocol it was trained for: two camera images and the robot state as input, five actions per query, 15 steps with the

arm at rest before the policy acts, at most 600 policy steps, and a stop at the first success. We run the initial states 0 to 9 of all 40 tasks, with the seed equal to the index of the initial state. The policy succeeds in 381 of 400 episodes (95.2%): 99 on LIBERO-Spatial, 97 on LIBERO-Object, 98 on LIBERO-Goal and 87 on LIBERO-10.

B.3 RoboCasa

We use RoboCasa (Nasiriany et al., 2024) on MuJoCo 3.3.1 with robosuite's split physics step, a 2 ms physics step and 25 physics steps per action, and XR-1 (Fan et al., 2025) with its RoboCasa checkpoint and evaluation protocol on ten object placement tasks, ten episodes each: Arrange Bread Basket, Coffee Setup Mug, Pick Place Counter To Cabinet, Pick Place Counter To Stove, Pick Place Drawer To Counter, Pick Place Sink To Counter, Pick Place Toaster To Counter, Stack Bowls Cabinet, Store Leftovers In Bowl and Wash Fruit Colander, and on the other 40 tasks of XR-1's evaluation set, five episodes each. The set holds 18 atomic tasks and 32 composite tasks, 16 of them unseen in XR-1's training; the ten placement tasks above are 4 atomic and 6 composite. XR-1 succeeds in 72 of the first 100 episodes and in 113 of the other 200.

B.4 RoboDojo

RoboDojo (Chen et al., 2026) runs Isaac Sim 5.1.0 and Isaac Lab 0.54.3 with two ARX X5 arms, a 4 ms physics step and 10 physics steps per action; RoboDojo switches PhysX to GPU dynamics. We run XR-1 with its official RoboDojo checkpoint on four tasks, Fill Pen Holder (three layouts), Insert Tubes (three), Put Bottles Into Dustbin (two) and Store Tools In Toolbox (two), under the three evaluation seeds that ship with the benchmark. Seed 0 was run twice, first with every episode replayed and then, together with seeds 1 and 2, without replay. XR-1 succeeds in 16 of the 40 episodes.

B.5 Detector parameters

Table 7: **Detector parameters** (Section 3.3), identical for all tasks and engines. The thresholds on velocity change are given for a 4 ms step and scale with the physics step on other engines.

Stage	Parameter	Value
Contact ejection	recent contact window	40 ms
	minimum speed	0.5 m/s
	minimum velocity change in one 4 ms step	0.35 m/s
	confirmation window	240 ms
	minimum displacement	4 cm
	free flight	$\geq$ 16 ms
	or violent contact with the container: velocity change / speed	$\geq$ 0.75 / 1.0 m/s
Gravity	window before onset for the drop height Δz	0.6 s
	window after onset for the peak speed	0.1 s
	ratio of peak speed to $\sqrt{2g\,\Delta z}$	> 1.5
	speed floor in the denominator	0.05 m/s
Carrier	window before onset for carriers	40 ms
	window after onset for the peak speed	0.1 s
	ratio of peak speed to the fastest carrier's speed	≥ 1.5
	speed floor in the denominator	0.05 m/s

C Reliability of the Tool

C.1 Recording, replay and reruns

Recording. SimuGuard wraps the physics step of the running scene, which leaves the evaluator, the benchmark code and the policy untouched. In SAPIEN, it records the drive targets, drive velocity targets and joint forces of every articulation before each step, 85 to 105 bytes per step; monitoring costs 7.5 to 11.6 ms per step on top of about 0.5 ms of physics. In MuJoCo, it records the control

inputs, applied forces and mocap poses, the solver's warm start whenever benchmark code changed it since the last step, and every write of positions, velocities or actuator states between two steps. For LIBERO, this takes 355 bytes per step for the actuation and 304 bytes for the object states, and monitoring adds 2.8 ms per step, which lengthens the wall time by 23% with cameras rendered on the CPU. In Isaac Sim, it records the drive position and velocity targets and joint forces of every articulation, reads states from PhysX tensor views and contacts from the PhysX contact report, which RoboDojo's object loaders are patched to enable; monitoring costs 6.9 to 18.4 ms per step, 3 to 7% of an XR-1 episode's wall time. Body velocities are those of the centers of mass.

Replay. A replay rebuilds the scene from its seed in a fresh process and applies the recorded actuation from the first physics step. In RoboTwin, it also restores the commanded gripper opening at each step from the gripper drive targets with RoboTwin's own mapping, because the success checks read it. In MuJoCo, it writes back the recorded warm start and the recorded writes between steps, and compares the full integration state with snapshots archived every 250 steps. On robosuite 1.4.0, LIBERO places fixtures with an unseeded random generator, so a scene is not reproducible from its seed; SIMUGUARD then writes the recorded body poses back before the replay. In Isaac Sim, restoring a saved state in place leaves trajectories 0.2 to 1 mm apart after 500 steps because PhysX's contact caches cannot be restored, while rebuilding the scene and replaying the recorded actuation is exact.

Reruns. The depenetration limit is set on every dynamic body and articulation link of the scene and stays on until the episode ends. A rerun from before the artifact switches it on at the physics step 0.1 s (25 steps) before the onset of the episode's first contact artifact and compares every recorded state before that step with the recording.

Replay checks, detection checks and recording cost. Table 8 collects every replay check we ran, the detection checks against human judgment, and the cost of recording on each engine. Restoring a saved state in place is not exact on either PhysX platform; rebuilding the scene and replaying the recorded actuation is.

Table 8: **Reliability of SIMUGUARD and cost of recording.** *Exact*: every recorded float of every tracked body matches the recording.

Check	Platform	Result
Replay		
Fresh process, 96 failures (2.42 M steps); events and outcomes reproduced	RoboTwin	96/96 exact; 43/43, 96/96
Stack Bowls Three, 12 failures: outcomes with / without gripper restore	RoboTwin	12/12 / 7/12
Another machine, 25 episodes	RoboTwin	25/25 exact
Prefix before a rerun switch, 273 replays	RoboTwin	273/273 exact
Full episodes, 400 (1.88 M steps); without the recorded warm start	LIBERO	400/400 exact; 0/33
Rebuilt scene, 13 episodes	RoboDojo	13/13 exact
Monitor on versus off	RoboTwin, RoboDojo	59 and 26 bodies identical
State restored in place during contact	SAPIEN / Isaac Sim	mm to cm / 0.2–1 mm apart
Detection		
Detected events judged implausible (Place Can Basket)	RoboTwin	37/40
Hard negatives judged implausible (Place Can Basket)	RoboTwin	0/20
Agreement on 60 episodes of six tasks, with / without the gravity stage	RoboTwin	58/60 / 39/60
Recording cost per physics step	SAPIEN	7.5–11.6 ms, 85–105 bytes
	MuJoCo (LIBERO)	2.8 ms, 659 bytes, +23% wall time
	Isaac Sim	6.9–18.4 ms, 3–7% of episode wall time

C.2 WHAT EACH DETECTION STAGE REMOVES

Table 9 counts, per platform, the events that the first stage flags, those that survive the gravity stage, and those that survive the carrier stage. The gravity stage does most of the work where objects are dropped: on RoboTwin's Put Bottles Dustbin (10 episodes) it removes all 20 flagged drops, and on the 60 human-labeled episodes of six further RoboTwin tasks it raises agreement with the annotators from 39 to 58 (Table 8). The carrier stage changes no label on RoboTwin and RoboDojo and removes none of RoboCasa's remaining events.

Table 9: **Detection stages across platforms.** Events flagged by the contact ejection stage, and those left after the gravity and carrier stages; episodes in parentheses where counted.

Platform	Episodes	Flagged	After gravity	After carrier
RoboTwin, Put Bottles Dustbin (LingBot-VA)	10	20	0	0
RoboTwin, 40 tasks, second pass (LingBot-VA)	800	12 (12)	10 (10)	10 (10)
RoboTwin, eight tasks, second pass (LingBot-VA)	200	169 (58)	22 (10)	17 (10)
RoboCasa, ten placement tasks (XR-1)	100	7 (7)	3 (3)	3 (3)
RoboCasa, the other 40 tasks (XR-1)	200	15 (15)	2 (2)	2 (2)
RoboDojo, four tasks (XR-1)	40	25 (15)	12 (7)	12 (7)
LIBERO, stored initial states, no policy	2000	655	650	650

D ROBOTWIN: CAUSE, STRESS TEST AND RERUNS

D.1 CAUSE IN THE COLLISION DETECTION

The reported contact. In the episode of Figure 1, replayed bit for bit, the pair that reports the 27.5 mm overlap at the onset step is one convex piece of the can and a 12 by 20 by 15 mm piece of the basket wall whose hulls are 21 mm apart when the step begins, beyond the two contact offsets of 10 mm, and absent from the contact report one step earlier. For the first artifact of each of the 37 Place Can Basket episodes that contain one, we take the deepest contact between can and basket at the onset step and compute the distance between the convex hulls of the two pieces that report it, at the poses the step starts from. In 35 episodes this contact is deeper than 5 mm. The two pieces are then a median 15.0 mm apart, more than 2 mm in 33 and more than 20 mm in 7. PhysX reports a normal that points from the second shape to the first; in the 1958 contacts between can and basket one step before these onsets, the can's piece lies on that side of the basket's piece in every case, and in the 35 onset contacts it lies on the other side in 34. The reported separation equals the distance, along the reported normal, from the vertex of the can's piece farthest from the basket to the contact point on the basket (median difference below 0.01 mm): the gap plus the thickness of the piece. One step earlier, the same pair reports a positive separation of 0.5 to 20 mm in 21 episodes and nothing in the others.

The path in PhysX 5.3.1. PhysX computes the contacts of two convex pieces from their closest points, found by GJK (Gilbert et al., 1988). At each iteration GJK adds a support point to a simplex and reduces it to the feature closest to the origin. In the reduction of a triangle abc, PhysX 5.3.1 treats the triangle as degenerate and falls back to a segment when $|ab \times ac|^2 \leq \epsilon$, with $\epsilon = 2^{-23} \approx 1.19 \times 10^{-7}$ (the single precision machine epsilon) and lengths in meters. The comparison does not scale with the geometry: a right triangle with legs of 1.8 cm is already below it. At the onset steps of four Place Can Basket episodes, the triangle that GJK builds has $|ab \times ac|^2$ of 8.81×10^{-8}, 1.189×10^{-7}, 1.180×10^{-7} and 5.99×10^{-8}, with edges of 11 to 96 mm. The triangle is rejected, the distance does not decrease, and the iteration ends through its exit for a degenerate simplex, with closest points that are not the closest and without the comparison against the contact distance that its regular exit makes; in the episode of Figure 1 it returns 26.95 mm for pieces 21.1 mm apart. The contact generation accepts this result when the search direction and the normal agree, and builds the full set of contact points from the faces on which the two closest points lie. The point on one piece lies inside it or on its far side, so the face that is selected points away from the other piece; the contact normal is taken from that face, and the separation of each point is measured to its plane, on the far side of which the other piece lies. The result is a negative separation of the gap plus the thickness of the piece. The points are reported whenever their separation is below the contact offset,

with no lower bound, and the solver removes the depth within the step. PhysX 5.4.0 compares with $\epsilon^2 \approx 1.42 \times 10^{-14}$; this is the only change to the simplex reduction between the two versions, and the steps after it are the same up to 5.9.0.

Reproduction outside the simulator. A program of about one hundred lines builds the two convex pieces from their vertices, calls PhysX's contact generation for one pair with a contact cache that the program keeps, and feeds it the poses of the two bodies at every step of the recording from the step the pieces first come within range. For four Place Can Basket episodes it returns, at the onset step, the normal and the separations that the simulator reported, to the printed digits (Table 10). With only the header of the simplex reduction taken from 5.4.0, the contact is not reported in any of the four; with only the other change that 5.4.0 makes to this contact routine, a corrected extent of the second hull, it is reported in all four. In the episode of Figure 1 the false contact needs the simplex that GJK stored in the cache five steps earlier and does not appear from an empty cache; in two of the four it appears from an empty cache as well. In the simulator, a pose written back unchanged to the can just before the onset, which clears the cache of its pairs, removes the artifact in 6 of 9 episodes.

On Stack Bowls Three we traced the 21 onset contacts deeper than 5 mm between two bowls in the 12 episodes with an artifact. A separating axis test on the two pieces gives the true distance. In 16 of them the pieces do not intersect (gaps of 0 to 21.7 mm) and the reported depth of 6 to 58 mm comes from the path above; the face that points away is on the first piece in 4, on the second in 7, and in 5 the reference face is the right one and the face paired with it on the other piece is nearly perpendicular to the normal. In 4, all late in one episode, the pieces do intersect, GJK reports an intersection, and the depth that the expanding polytope algorithm returns equals the true one (11.7 to 18.0 mm). One is not reproduced by the program. With the threshold of 5.4.0, none of the 16 is reported and the 4 real penetrations keep their depth.

Table 10: **The false contact across versions of PhysX.** Onset contacts returned by PhysX's contact generation outside the simulator for the recorded poses: 4 false contacts on Place Can Basket, 16 false contacts and 4 real penetrations on Stack Bowls Three. A dash: not run.

Version	Can, false (4)	Bowls, false (16)	Bowls, real (4)
5.3.1	4	16	4
5.3.1 with the simplex reduction of 5.4.0	0	0	4
5.3.1 with the hull extent of 5.4.0	4	–	–
5.4.0, 5.4.2, 5.5.1, 5.6.1	0	–	–
5.9.0	0	0	4

Inside the simulator. SAPIEN ships PhysX inside its own library, and the releases 3.0.0 and 3.0.3 still contain 5.3.1. We copy the library of SAPIEN 3.0.0b1 and change, in the copy, the constant that the four triangle tests of the simplex reduction load, from 2^{-23} to 2^{-46}; 19 bytes differ, and nothing else in the engine changes. With the unchanged copy, the episode of Figure 1 replays bit for bit over 60922 steps. With the repaired copy, the recorded actions of four Place Can Basket episodes give no contact between can and basket deeper than 5 mm and a peak can speed below 1 m/s, against 27.5 mm and 21.8 m/s in the episode of Figure 1. These replays leave the recording between steps 3499 and 5322, long before the artifact near step 11000, because the threshold also enters regular contacts during the grasp; an open loop replay in the repaired engine is therefore not paired with the recording, and the repaired engine is evaluated with the scripted expert (Table 2a) and in closed loop (Table 3). For the scripted expert we run RoboTwin's demonstration on seeds 100000 to 100059 under the monitor in both engines. On Place Can Basket, the plan succeeds on 57 and 60 seeds and the demonstration on 51 and 59; on Stack Bowls Three, on 35 and 36, and on 30 and 36. Of the seeds on which both engines finish, 8 and 6 succeed only in the repaired engine, and 0 and 1 only in the benchmark's.

Contact offset. A shape generates contacts with every shape closer than the sum of their contact offsets, 10 mm each in SAPIEN. With the offset of every shape of the task's objects and containers changed 0.1 s before the first artifact, the Place Can Basket episodes that keep an artifact grow with the offset (Table 11); all 37 replays match their recordings up to the switch at every value. On Stack

Bowls Three at 1 mm, 6 of the 12 episodes keep an artifact and 4 failures pass: bowls that rest in each other have pieces closer than any offset.

Table 11: **Contact offset.** The 37 Place Can Basket episodes of LingBot-VA with an artifact (152 events, 21 failures, 16 successes), replayed with the contact offset of can and basket changed 0.1 s before the first artifact. The last row is the recording.

Contact offset	Episodes with an artifact	Events	Failures that pass	Successes kept
0.5 mm	3	10	18/21	10/16
1 mm	4	14	19/21	12/16
2 mm	9	29	18/21	14/16
3 mm	11	25	18/21	12/16
5 mm	13	46	16/21	10/16
10 mm	37	152	–	–

Closed loop evaluation in the repaired engine. Table 3 evaluates LingBot-VA on 20 scored seeds per task in four arms: RoboTwin as released; the contact offset of objects and containers at 1 mm, set at the first action of the policy so that the scripted demonstration runs under the released settings; the repaired engine, in which the demonstration runs as well; and the persistent contact manifold switched off for the whole scene. LingBot-VA draws its sampling noise without a seed, and in these runs the noise generator is reseeded at every reset, so every episode sees the same noise sequence. The scripted demonstration is not deterministic across runs, so the arms do not score exactly the same seeds, and paired counts are over the seeds scored in both arms. Against the released benchmark on the five container tasks: the repaired engine shares 83 seeds, 73 succeed in both, 8 only in the repaired engine and 2 only in the released one; the offset arm shares 85 seeds, with 74, 8 and 2, and one failing in both. Table 12 lists all four arms on the seven tasks. With the manifold off, 97 of 100 episodes on the five container tasks succeed, but 3 still contain an artifact, 2 of them failures; the repaired engine has none. On the two tasks without a container, the arms differ by one to three episodes with no artifact in any of them, which is the spread between runs of 20 episodes here. HyVLA in the repaired engine (Table 3) is compared with its audited run, which was not reseeded; the two share 83 scored seeds, of which 7 succeed only in the repaired engine and 4 only in the audited run. Its one episode with an artifact in the repaired engine, on Place Bread Basket, is a bread that leaves the other bread at 2.4 m/s, 21 times the other bread's speed, and the scripted demonstration on that seed shows the same between the two breads; the pieces involved are not those of a container wall, and we have not traced this case.

Table 12: **Closed loop evaluation, all arms.** LingBot-VA, 20 scored seeds per task, sampling noise reseeded at every episode; successes, in parentheses the episodes with an artifact after the gravity stage, and the seeds the scripted demonstration failed on. *Manifold off*: PhysX's persistent contact manifold switched off for the whole scene.

Task	Benchmark	Offset 1 mm	Repaired	Manifold off
Place Can Basket	15 (5), 4	19 (3), 4	20 (0), 2	20 (0), 3
Stack Bowls Three	15 (6), 18	17 (3), 16	19 (0), 11	18 (2), 12
Place Object Basket	20 (0), 14	20 (0), 12	20 (0), 12	20 (1), 3
Place Bread Basket	19 (1), 6	19 (0), 5	20 (0), 4	20 (0), 3
Stack Bowls Two	20 (0), 7	20 (0), 7	19 (0), 7	19 (0), 7
Blocks Ranking Size	18 (0), 1	17 (0), 1	19 (0), 2	17 (0), 3
Open Microwave	15 (0), 20	18 (0), 20	16 (0), 20	17 (0), 37
Five container tasks	89 (12)	95 (6)	98 (0)	97 (3)

D.2 STRESS TEST

The scene contains RoboTwin's can (model 1 of object 071, 22 convex pieces, 10 g) and basket (model 1 of object 110, 54 convex pieces, 0.5 kg) on a table, with the same collision pieces, masses, inertias and center of mass frames in SAPIEN 3.0.0b1 with PhysX and in MuJoCo 3.3.1, gravity on

and a 4 ms step. The basket takes its pose from the episode of Figure 1 one physics step before the ejection. Each of the 300 can poses has a uniformly random orientation and a position within 3 cm of the recorded one horizontally and from 3 cm below to 1 cm above it vertically; the can starts at rest, and 288 poses start at least 5 mm deep in the basket. Each pose runs for 0.5 s (Table 13).

Table 13: **Stress test.** RoboTwin's can at 300 resting poses inside its basket, in PhysX and in MuJoCo with identical assets and step. *Thrown*: at least 1 m/s and 4 cm away. *Removal*: initial overlap over exit speed. RoboTwin's values in parentheses.

Engine and setting	Thrown (%)	Exit speed (m/s)	Distance (cm)	Removal (ms)
PhysX, RoboTwin settings	91	8.77	155	2.6
PhysX, can mass 100 g (10 g)	90	6.86	73	3.3
PhysX, solver iterations 32/8 (10/1)	90	16.78	200	1.4
PhysX, depenetration $\leq$ 1 m/s	65	1.36	10	16.5
PhysX, depenetration $\leq$ 0.1 m/s	0	0.14	3.7	169
MuJoCo, default (time constant 20 ms)	5	0.56	2.8	38.6
MuJoCo, time constant 8 ms	42	1.99	3.8	10.6

The stress test starts from real overlaps and measures how each engine removes them. Mass and solver iterations leave the share of thrown poses at 90%; only the maximum depenetration velocity, the parameter PhysX's documentation names for bodies that start in overlap (NVIDIA Corporation, 2024), changes it.

The response depends on depth. With the can pushed straight down into the basket floor, PhysX with RoboTwin's settings releases it at 0.04 m/s from depths up to 10 mm and at 4.20 m/s from 20 mm (peak speed within 0.1 s); with the depenetration limit at 1 and 0.1 m/s, the 20 mm case leaves at 1.24 and 0.14 m/s, and with 32 position iterations at 11.92 m/s. MuJoCo with its default contact releases the can at 0.24 m/s from 10 mm and at 1.05 m/s from 20 mm. MuJoCo does not accept a time constant below two physics steps; with this safeguard disabled, time constants of 4 or 1 ms make the same scene diverge. The recorded poses of can and basket one step before the ejection, placed in this scene without the robot, show no overlap and do not reproduce the ejection (0.23 m/s), consistent with the recording, in which the pieces do not intersect and the reported depth comes from the contact cache (Section D.1); the stress test therefore starts from real overlaps of the depth reported at onset rather than from the recorded state.

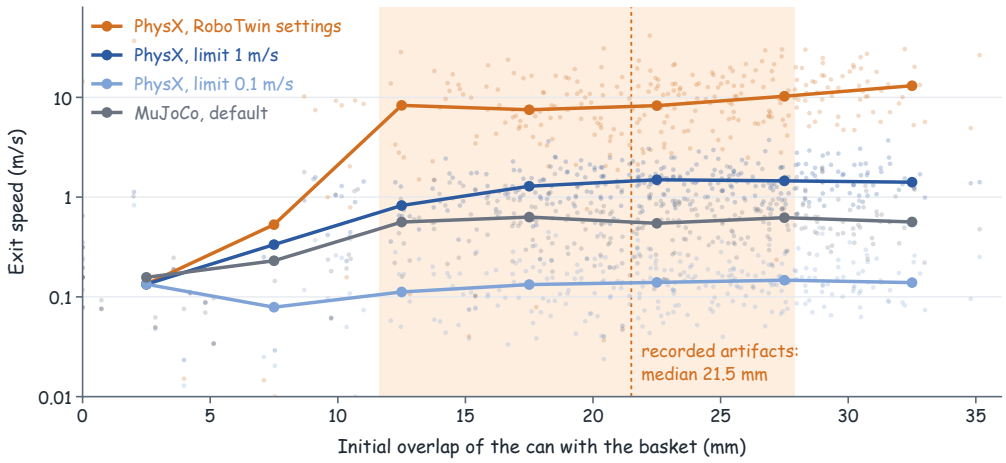


Figure 4: **Exit speed against initial overlap** in the stress test of Table 13: RoboTwin's can at 300 resting poses inside its basket, without a robot, in PhysX with RoboTwin's settings, with the depenetration limit at 1 and 0.1 m/s, and in MuJoCo with its default contact. Dots are poses; lines are medians in 5 mm bins. With RoboTwin's settings the median exit speed is 0.13 m/s below 10 mm of overlap and 9.4 m/s at or above it; the limits and MuJoCo stay between 0.14 and 1.5 m/s at every depth. The band marks the overlap at the onset of the 99 recorded artifacts on Place Can Basket, median 21.5 mm and interquartile range 11.6 to 27.9 mm, above the depth at which PhysX starts to eject.

D.3 Reruns

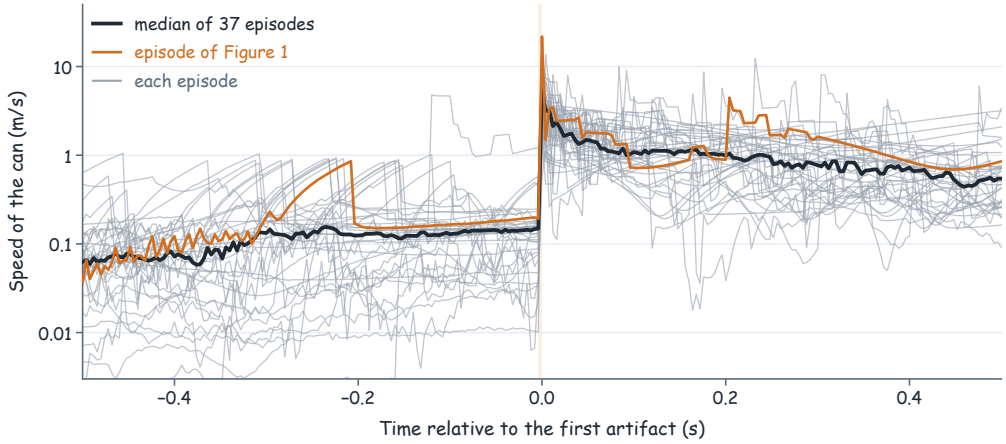


Figure 5: **The can's speed around the first artifact** in all 37 Place Can Basket episodes of LingBot-VA that contain one (21 failures, 16 successes), at every 4 ms step from 0.5 s before to 0.5 s after the onset. Gray: episodes; bold: median; orange: the episode of Figure 1. One step before the onset the can moves at a median 0.15 m/s. The largest one step rise in speed falls on the onset step in 33 episodes and within four steps in the other 4; it ranges from 0.53 to 21.6 m/s, median 5.7 m/s, more than an order of magnitude in the median.

Contact manifold. The warm start of GJK is stored in PhysX's persistent contact manifold, which RoboTwin leaves on. With the manifold switched off and nothing else changed, PhysX generates contacts by another routine, and the recorded actions replayed from the first step give an artifact in 2 of the 37 Place Can Basket episodes with an artifact (152 events fall to 3, the median peak

speed from 7.89 to 0.91 m/s) and in 1 of the 8 Stack Bowls Three episodes (83 events to 1, 5.22 to 1.31 m/s); the success check passes 27 and 6 of these open loop replays against 16 and 0 as recorded. These replays diverge from the recording at the first contact, so they are not paired with it.

Changes from the first physics step. Applied from the first step, each change of a simulation parameter also rescues Place Can Basket failures that contain no artifact (Table 14), so these replays do not separate the parameters. A clamp that caps the object's speed at 2 m/s after each step, which is not a simulation parameter, rescues none of them.

Table 14: **Replays from the first physics step** of the recorded failures: failures that pass the success check.

	Place Can Basket failures		Stack Bowls Three
Change	with artifact (21)	without artifact (2)	failures (12)
Depenetration $\leq$ 1 m/s	19	2	8
Depenetration $\leq$ 0.1 m/s	9	2	5
Object mass 100 g	17	2	6
Solver iterations 32/8	14	1	5
Speed clamp 2 m/s	14	0	0

One setting changed before the artifact. Switched on 0.1 s before the first artifact, with everything before identical to the recording, a smaller contact offset and the depenetration limit rescue the most failures (Table 15); a ten times heavier object and more solver iterations rescue fewer and leave the object faster.

Table 15: **Replays switched on 0.1 s before the first artifact.** LingBot-VA episodes with artifacts, replayed with one setting changed. *Rescued*: failures that pass. *Peak*: median object speed after the switch over all episodes with an artifact. The contact offset is changed on the task's objects and containers, the other settings as in Section 3.5.

	Place Can Basket		Stack Bowls Three	
Change	Rescued	Peak (m/s)	Rescued	Peak (m/s)
Contact offset 1 mm (10 mm)	19/21	0.91	4/12	2.50
Depenetration $\leq$ 1 m/s	18/21	1.39	7/12	0.65
Object mass 100 g	14/21	3.72	5/12	2.12
Solver iterations 32/8	12/21	11.20	4/12	7.00

Limit value. Switched on 0.1 s before the first artifact, limits from 0.1 to 1 m/s rescue nearly the same failures, and a limit of 2 m/s rescues fewer while the object leaves faster (Table 16). The peak speeds here are medians over the 21 failures; Table 15 takes the median over all 37 Place Can Basket episodes with an artifact, successes included, which is why it reads 1.39 m/s against 1.25 m/s here. All 108 replays of this sweep match their recordings up to the switch.

Table 16: **Limit value.** LingBot-VA failures with artifacts that pass when replayed with the depenetration limit switched on 0.1 s before the first artifact, and the median peak speed of the object after the switch over these failures. *Three tasks*: Place Object Basket, Place Bread Basket and Stack Bowls Two, one failure each. The successes with an artifact were replayed at 1 m/s only.

	Place Can Basket (21)		Stack Bowls Three (12)		Three tasks (3)	Five tasks
Limit	Rescued	Peak (m/s)	Rescued	Peak (m/s)	Rescued	Rescued
0.1 m/s	18	0.91	7	0.40	3	28/36
0.5 m/s	18	1.05	7	0.63	2	27/36
1 m/s	18	1.25	7	0.65	3	28/36
2 m/s	16	1.95	4	1.09	3	23/36

Successes with an artifact. The same replay is run for every success that contains an artifact; Table 22 lists how many keep their success per task and policy.

Closed loop evaluation with the limit on. We also evaluate LingBot-VA again with the depenetration limit of 1 m/s on for the whole policy rollout, with the scripted demonstration unchanged (Table 17). On Place Can Basket it succeeds in 81 of 90 episodes over three repetitions, against 67 reported; Stack Bowls Three and Stack Bowls Two rise to 20 of 20, Place Bread Basket falls from 19 to 17, and Place Object Basket from 18 to 10, with artifacts still present in 2 of its 20 episodes. The tasks without container artifacts move in both directions, from 15 to 10 of 20 on Open Microwave, from 7 to 5 of 10 on Put Object Cabinet, and from 17 to 18 on Blocks Ranking Size. The policy samples its actions without a fixed seed, so each closed loop episode is a new rollout, and 20 episodes resolve such differences only coarsely. Put Bottles Dustbin was not completed. The as run rate on the shared seeds counts each seed once with its recorded outcome. HyVLA, run the same way on the five container tasks, keeps 16 of 20 on Place Can Basket, rises to 20 of 20 on both bowl tasks, and falls from 20 to 14 of 20 on Place Bread Basket and from 19 of 20 to 10 of 22 on Place Object Basket (18 seeds, four of them run twice); 7 of its 102 episodes still contain an artifact.

Table 17: **Closed loop evaluation with the limit on.** Successes as reported and with the depenetration limit of 1 m/s on for the whole policy rollout. *The first 20 scored seeds of the reported run of 50.

Task	Reported	Limit on
LingBot-VA		
Place Can Basket	67/90	81/90
Stack Bowls Three	16/20*	20/20
Stack Bowls Two	19/20	20/20
Place Bread Basket	19/20	17/20
Place Object Basket	18/20	10/20
Stamp Seal	19/20	19/20
Blocks Ranking Size	17/20	18/20
Open Microwave	15/20	10/20
Turn Switch	9/20	9/20
Put Object Cabinet	7/10	5/10
Place Burger Fries	20/20	20/20
HyVLA		
Place Can Basket	16/20	16/20
Stack Bowls Three	18/20	20/20
Stack Bowls Two	17/20	20/20
Place Bread Basket	20/20	14/20
Place Object Basket	19/20	10/22

Closed loop evaluation with changed physics. Changing one parameter for the whole closed loop evaluation also changes the reported rate (Table 18). With a can of 100 g, LingBot-VA succeeds on Place Can Basket in 95.6% of episodes against 74.4% with RoboTwin's 10 g can (paired over seeds, $p = 0.0015$; 95% interval of the difference $[+10.3, +32.8]$ points). More solver iterations give 86.7% (Wilcoxon signed rank test on success per seed, $p = 0.17$).

Table 18: **Closed loop success** of LingBot-VA on Place Can Basket under three physics configurations, 30 scored seeds per repetition. The solver configuration completed two repetitions.

Configuration	Rep. 1	Rep. 2	Rep. 3	Total	Mean $\pm$ std (%)
RoboTwin (can 10 g)	23/30	24/30	20/30	67/90	74.4 ± 6.9
Can mass 100 g	29/30	30/30	27/30	86/90	95.6 ± 5.1
Solver position iterations 32	26/30	26/30	–	52/60	86.7 ± 0.0

Closed loop takeover. We also let the policy take over from a replayed prefix, 1 s before the first artifact of each of the 21 Place Can Basket failures, with the speed clamp on or with the physics

unchanged. All 42 prefixes match their recordings bit for bit. The unchanged physics rescues 10 of 21 failures and the clamp 15 (paired: 7 rescued only with the clamp, 2 only without; McNemar exact test $p = 0.18$). From the same state, the policy's own sampling often avoids the artifact, which is why the corrected rate of Equation (1) replays the recorded actions instead.

Skipped scripted demonstrations. RoboTwin scores a policy on a seed only if its own scripted demonstration passes there first. On Place Can Basket, 22 seeds were skipped for this reason over the three repetitions, and the demonstration of 16 of them contains a contact artifact. Replayed with the depenetration limit switched on 0.1 s before the first artifact, 13 of the 16 pass the success check, with all 16 prefixes identical to the recording up to the switch (Figure 3, bottom). The artifacts therefore also decide which seeds enter the evaluation, before any policy is scored (Table 19).

Table 19: **The benchmark's own expert fails the same way.** Place Can Basket: failures that contain an ejection, and failures that pass the success check when replayed with the depenetration limit switched on before it. The scripted expert acts on ground truth object poses with a motion planner; RoboTwin runs it on every seed before scoring a policy and skips the seed when it fails.

Actor	Episodes	Failures	With an ejection	Pass with the limit
LingBot-VA (world action model)	90	23	21	18
HyVLA (vision language action)	20	4	1	1
Scripted expert (ground truth poses)	22	22	16	13

E Other Platforms

The adapters for RoboDojo, LIBERO and RoboCasa are described in Section C.1; this section reports what the audit found on each.

E.1 RoboDojo

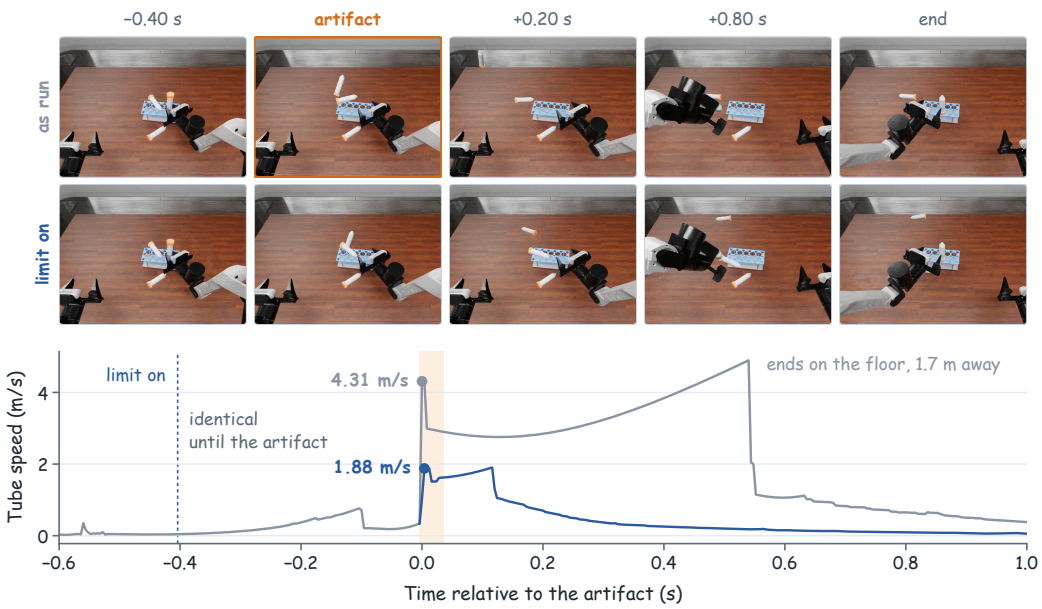


Figure 6: **RoboDojo, Insert Tubes (layout 2), XR-1.** Head camera frames at fixed times relative to the first artifact, as run (top) and replayed with the depenetration limit of 1 m/s switched on 0.4 s before it (bottom), and the speed of the ejected tube in both runs. As run, the tube leaves the rack at 4.31 m/s within one step, flies off the table and lands 1.7 m away; the later peak of the gray curve is its fall. With the limit it leaves at 1.88 m/s, slows below 0.4 m/s within 0.3 s and stays on the table. Both rows are replays of the recording; the as run replay matches it bit for bit, and the limited one first differs at the artifact step.

Seven of the 40 XR-1 episodes contain contact artifacts, all failures. In the first pass over seed 0: two events on Insert Tubes (layout 2) and three on Store Tools In Toolbox (layout 1). In the later passes: two on Insert Tubes (layout 2 under seed 0, at 6.30 m/s, and layout 0 under seed 2), four on Fill Pen Holder (layouts 1 and 2 under seed 0) and one on Store Tools In Toolbox (layout 1 under seed 1). The first detection stage confirmed 25 events in 15 episodes, and the gravity stage explained 13 of them as falls. Only the first pass was replayed. We replayed the Insert Tubes episode with the depenetration limit switched on 100 physics steps (0.4 s) before the ejection; the replay is identical to the recording up to the ejection itself (Table 20).

The contact that starts this ejection is false as well. RoboDojo runs Kit 107.3 with GPU dynamics; the tube is a dynamic body of 60 g with a convex decomposition, the rack a static collider approximated by a signed distance field, and the tube's depenetration limit reads 3 m/s at run time. The gripper releases the tube about 120 steps before the onset, after which no robot link touches it, and the tube tips over the upper edge of the rack. At the onset step three new contact points report separations of -10.9 to -10.1 mm with an impulse of 1.08 N s at the deepest. They lie on the side of a wall of the rack that is 2 to 5 mm thick, and 7.6 mm outside the convex hull of the tube's mesh, which contains every piece of its decomposition; the point on the tube that each implies lies on the other side of that wall. In the 14 steps before, that point of the rack is 6.8 to 7.6 mm from the tube's hull with no contact reported, and the tube moves 0.9 mm there in the last step. The engine therefore pairs the tube with the far side of a thin wall and reports the gap plus the wall as a depth, as on RoboTwin, but between convex pieces and a signed distance field in the GPU contact generation, where the threshold of Section D.1 plays no part; we have not located the faulty step there.

Table 20: **RoboDojo, Insert Tubes layout 2**: speed at which the tube leaves, replayed with the depenetration limit switched on 0.4 s before the ejection.

Depenetration limit	Exit speed (m/s)
As run, 3 m/s by default (replay identical to the recording)	4.31
1 m/s	1.88
0.3 m/s	0.62

E.2 LIBERO

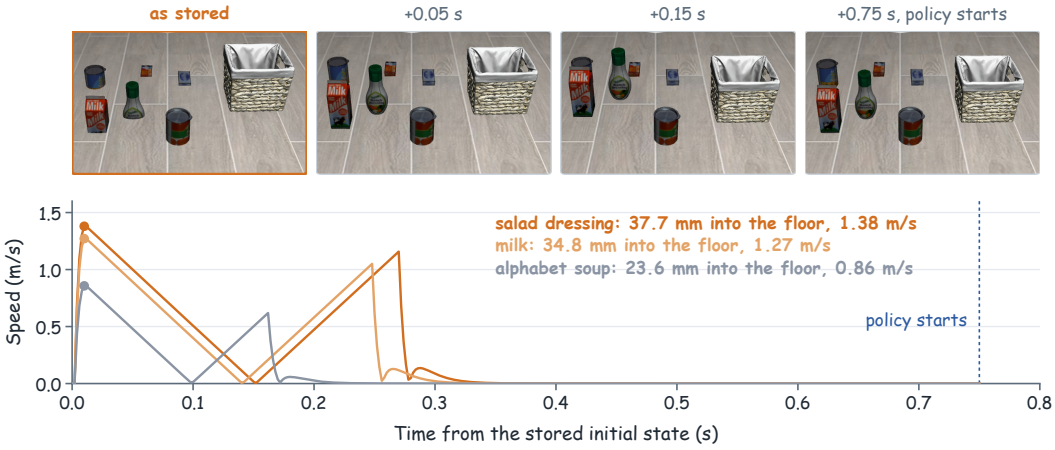


Figure 7: **LIBERO-Object task 0, stored initial state 0**, built as the evaluation builds it (official LIBERO, robosuite 1.4.0, MuJoCo 3.2.3) and run through the 15 settle steps that precede every LIBERO policy step. As stored, the salad dressing, the milk and the alphabet soup sit 37.7, 34.8 and 23.6 mm inside the floor. MuJoCo pushes them out within 10 ms at 1.38, 1.27 and 0.86 m/s; they hop, land within 0.3 s and are at rest before the policy acts at 0.75 s. Depths and peak speeds match the audit of all 2000 stored states, of which 650 start inside a support.

Table 21: **LIBERO's stored initial states**, built as the evaluation builds them and simulated for the 15 steps before the policy acts, without a policy.

Suite	Initial states	Contact ejection	After gravity	Contact artifacts	Tasks affected
LIBERO-Spatial	500	3	0	0	0/10
LIBERO-Object	500	500	500	500	10/10
LIBERO-Goal	500	0	0	0	0/10
LIBERO-10	500	152	150	150	3/10

Initial states. In the floor scene of LIBERO-Object and the living room table of LIBERO-10 tasks 0, 1 and 7, the stored initial states place objects inside their support (Table 21): the salad dressing 37.7 mm, the ketchup 37.4 mm and the milk 34.8 mm into the floor, and the ketchup 14.6 mm into the table. The depth is the same in every initial state of a scene, which differ only in horizontal positions. MuJoCo removes the overlap within the first 10 ms by launching the objects, at up to 1.38 m/s off the floor and 1.83 m/s off the table, with no robot contact. In the 130 policy episodes that start from these states, the policy succeeds in 96.2%, against 94.8% in the other 270.

Episodes with the policy. Two of the 400 episodes contain a contact artifact while the policy acts. In LIBERO-10 task 9 (a mug into the microwave), MuJoCo reports a contact 242.8 mm deep with 412 N between a collision box of the mug and a plate of the microwave, although its own distance query gives 1.4 mm between the two shapes; the mug leaves the gripper at 1.57 m/s and the episode fails. Whether this contact caused the failure is not tested. In LIBERO-Object task 4, a finger

touches the ketchup again during the release, and its speed rises from 0.25 to 0.62 m/s for one step; the episode succeeds.

E.3 RoboCasa

Five of the 300 XR-1 episodes contain a contact artifact, all of them failures. On the ten placement tasks: bread against its basket at 0.61 m/s on Arrange Bread Basket, an object leaving a drawer at 1.19 m/s on Pick Place Drawer To Counter, and a bowl on the floor at 0.53 m/s on Stack Bowls Cabinet. On the other 40 tasks, which include doors, drawers, appliances and long horizon composite tasks: a baguette against the door of a toaster oven at 0.61 m/s on Heat Kebab Sandwich, and a piece of meat on the floor, pushed by the mobile base, at up to 0.84 m/s on Pack Identical Lunches. The first detection stage flags 22 of the 300 episodes; the gravity stage explains the events in the other 17 as falls.

F Episode Accounting

Table 22 gives the counts behind Table 1: failures with an artifact out of all failures, the failures that pass when replayed with the limit, and the successes with an artifact that fail. It also lists the tasks outside the five, which Table 1 omits because their audit is incomplete: HyVLA's successes on its six other tasks were not checked for artifacts, HyVLA was not run on the remaining 39 tasks, and the closed loop runs cover the five tasks and, for LingBot-VA, the six others (Table 17), not the remaining 39.

Table 22: **Counts behind Table 1.** *Art.*: failures with an artifact, of all failures. *Resc.*: failures that pass under the limit. *Lost*: successes that fail under the limit, of all successes with an artifact. †Both passes of every task (Table 5); of the 40 tasks only the first pass, 4 failures and 3 successes with an artifact, was replayed, and the three Put Object Cabinet replays of the second pass diverge from the recording before the switch and are not counted.

	LingBot-VA			HyVLA		
Task	Art.	Resc.	Lost	Art.	Resc.	Lost
Place Can Basket	21/23	18	7/16	1/4	1	2/3
Stack Bowls Three	12/12	7	–	2/2	1	0/1
Place Object Basket	3/5	3	2/3	0/1	–	–
Place Bread Basket	1/1	1	–	0/0	–	–
Stack Bowls Two	3/3	3	–	3/3	3	–
Five tasks	40/44	32	9/19	6/10	5	2/4
Six other tasks	1/24	1	–	0/38	–	–
Other 45 tasks†	12/146	1	1/9	–	–	–
Rates outside the five tasks (as run / replay / exclusion), LingBot-VA then HyVLA						
Six other tasks	76.0 / 77.0 / 76.8			68.3 / 68.3 / –		
Other 45 tasks†	91.9 / – / –			– / – / –		

Table 23 lists every scored episode that contains a contact artifact, with its seed, recorded outcome and the outcome of its replay with the depenetration limit of 1 m/s switched on 0.1 s before the first artifact. Equation (1) sums $r_e - y_e$ over these rows. Place Can Basket was evaluated in three repetitions of 30 seeds (Section B), so a seed can appear in more than one repetition.

Table 23: **Every scored episode with a contact artifact** and its replay. S: success, F: failure; the arrow leads from the recorded outcome y_e to the replayed outcome r_e.

Task	Episodes	Seed, recorded → replayed
LingBot-VA		
Place Can Basket (rep. 1)	13	100000 F→S, 100002 S→S, 100004 S→F, 100006 F→S, 100007 F→S, 100008 F→S, 100014 S→S, 100017 S→F, 100021 F→S, 100022 S→S, 100024 S→F, 100029 F→S, 100031 F→F
Place Can Basket (rep. 2)	11	100004 S→S, 100011 F→S, 100012 F→S, 100017 S→F, 100019 F→S, 100021 S→S, 100022 F→S, 100025 S→F, 100026 S→S, 100027 S→F, 100029 F→F
Place Can Basket (rep. 3)	13	100000 S→S, 100003 F→S, 100009 F→S, 100016 F→F, 100019 F→S, 100021 F→S, 100022 F→S, 100023 F→S, 100027 F→S, 100028 F→S, 100029 S→F, 100030 S→S, 100032 S→S
Stack Bowls Three	12	100002 F→S, 100009 F→F, 100023 F→F, 100028 F→F, 100047 F→S, 100067 F→S, 100071 F→S, 100073 F→F, 100075 F→S, 100077 F→S, 100078 F→S, 100089 F→F
Place Object Basket	6	100013 F→S, 100015 S→S; pass 2: 100001 F→S, 100009 S→F, 100014 S→F, 100016 F→S
Place Bread Basket	1	100003 F→S
Stack Bowls Two	3	100011 F→S; pass 2: 100014 F→S, 100027 F→S
Dump Bin Bigbin	1	pass 2: 100010 F→F
Put Object Cabinet	3	pass 2: 100110 S→S, 100129 S→S, 100210 F→S (replays diverge before the switch; not counted)
Place Fan	1	100015 F→F
Hanging Mug	1	100003 F→F
Stamp Seal	1	100026 F→S
Move Stapler Pad	1	100018 F→F
Place Burger Fries	2	100012 S→S, 100015 S→S
Place Dual Shoes	1	100015 S→F
HyVLA		
Place Can Basket	4	100004 S→F, 100021 S→F, 100022 F→S, 100023 S→S
Stack Bowls Three	3	100016 S→S, 100022 F→S, 100023 F→F
Stack Bowls Two	3	100020 F→S, 100021 F→S, 100026 F→S

G HUMAN VERIFICATION PROTOCOL

For Place Can Basket, 60 clips were rendered by replaying recorded episodes: 40 centered on a detected event and 20 hard negatives (the fastest object motion in episodes without an event, and detector flags that were never confirmed). Clips were shuffled and shown without their origin to five robotics engineers, who rated them independently and then resolved disagreements by discussion; the labels reported are the agreed ones. For each clip, they answered whether any object moves in a physically implausible way, and whether the robot was behaving sensibly for the task before that moment; the robot was judged on track in 38 of the 40 events. For six further tasks, 60 complete evaluation videos were shown in random order without seed or outcome, with three questions: implausible motion anywhere in the episode (objects falling after release excluded), sensible robot behavior, and whether the task was completed. Their completion judgments matched the benchmark's in 59 of 60 videos.